

NEW JERSEY DEPARTMENT OF TRANSPORTATION

E-Permitting System Customer Portal Reference Guide

COMPLETE BUSINESS, FUNCTIONAL, WORKFLOW, & AUTOMATION KNOWLEDGE BASE

A professional, reverse-engineered solution guide detailing the NJDOT E-Permitting Customer Portal, covering highway occupancy, utility openings, drainage connections, subdivision planning, and automated testing architectures.

Prepared for Developers, Business Analysts, Product Owners, UI/UX Teams, QA Engineers, and Automation Teams.

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Author	Senior Solution & QA Architect (Advanced Agentic Systems)

1. Executive Summary

The New Jersey Department of Transportation (NJDOT) manages all physical encroachments, access links, utility lines, and structural connections along State highways. The NJDOT E-Permitting System acts as the customer-facing web application where companies, utility providers, developers, and independent applicants file digital requests.

By digitizing these workflows, NJDOT enforces statewide safety specifications, tracks physical access points, automates the collection of statutory fees, and coordinates field engineering reviews. This reference document acts as a complete functional and technical knowledge base to support product owners, developers, and QA teams.

2. Module Overview

The E-Permitting Customer Portal includes several integrated functional modules:

- Profile Registry: Captures organization details, tax identifiers, and Point of Contact (POC) records.
- Dashboard: Main portal containing the 'Submit Application' launcher card.
- Submit Application drawer: Portal dispatch system supporting 12 active permit categories.
- Action Items Grid: Tracks pending revisions or supplementary files requested by NJDOT engineers.
- Permit History Grid: Archives historical submissions and their status logs.
- Payment Activity Ledger: Financial ledger showing paid fees, transactions, and Excel download exports.
- Common Checkout Gateway: Renders the credit/debit card billing gateway.

3. Business Purpose

The e-permitting portal serves several critical state operational requirements:

- Statutory Enforcement: Integrates state regulations for highway utility openings and driveway access (NJSA 27:7-44.1 and NJAC 16:41).
- Safety Assurance: Reviews structural load capabilities (bridge attachments), utility depths, and line-of-sight clearances (poles and structures).
- Administrative Efficiency: Standardizes review metrics, replaces physical files with structured data, and coordinates multi-agency approvals.

4. User Roles & Permissions

The application divides authorization into the following roles:

- Utility Provider / Contractor: Submits utility openings, pole erections, or bridge attachments, and handles billing.
- Land Developer: Prepares lot consolidation, major planning, and driveway access requests.
- Appointed Engineering Agent: Represents land developers or utility companies during the planning review.
- NJDOT Engineer / inspector (Staff Portal): Backend staff who review designs, execute coordinate inspections, issue permits, and audit financial files.

5. Functional Overview

The system coordinates data and document collections across 12 distinct permit types:

- Utility Opening: Applies to excavation, pipeline installation, or road surface cuts.
- Drainage Connection: Standardizes runoff volume estimations and drainage pipe layouts.
- Erection of Pole: Manages safety and placement spacing for utility poles.
- Highway Occupancy: Governs staging of construction materials or temporary highway detours.
- Bridge Attachment: Collects structural plans for mounting conduits or utility cables on state bridges.
- Letter of Interest: Initiates preliminary inquiries regarding highway access permits.
- Lot Consolidation/Subdivision: Manages parcel mapping and municipal approvals.
- Major Planning & Driveway Permits: Governs large-scale residential and commercial development access.
- Pre-Application Meeting: Requests face-to-face planning discussions with NJDOT staff.
- Street Intersection: Regulates physical intersections connecting local roads to state highways.

6. End-to-End Workflows

Workflow 1: Utility Road Opening Permit

- Step 1: User clicks 'Submit Application' on dashboard and selects 'Utility Opening'.
- Step 2: User inputs Lot Owner details (Company, POC name, phone, email, and address).
- Step 3: User uploads the required Authorization Certificate.
- Step 4: User selects Route, Suffix, Direction, and enters Milepost, Block, and Lot details.
- Step 5: User clicks 'Add New' to enter road cut dimensions (Length, Width, Depth).
- Step 6: System automatically calculates area (Length x Width) and validates that the result is greater than zero.
- Step 7: User uploads Overall Site Plans and checks the statutory acknowledgment box.
- Step 8: User clicks 'Continue to Payment', completes checkout, and returns home.

Workflow 2: Drainage Connection Permit

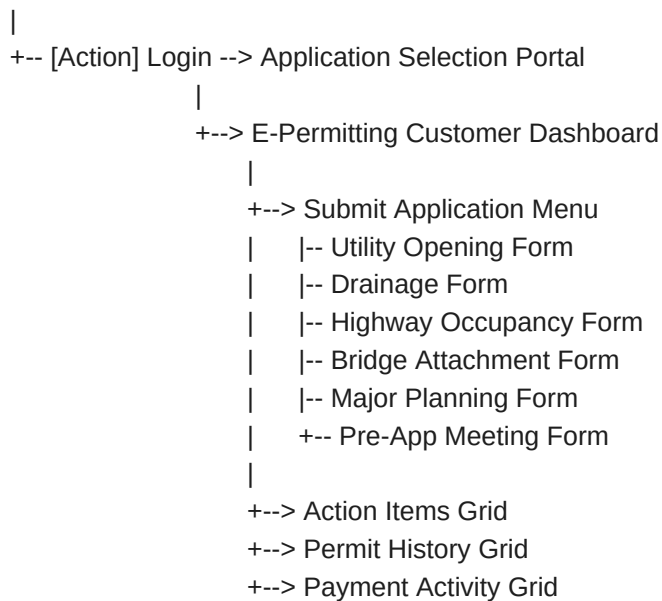
- Step 1: User launches Drainage form.
- Step 2: Inputs owner details and adds highway route location details.
- Step 3: Enters drainage pipe size, runoff coefficient, and rainfall intensity metrics.
- Step 4: Uploads Hydrology and Hydraulic calculations alongside Overall Site Plans.
- Step 5: Signee inputs name and title, accepts terms, and processes payment.

Workflow 3: Pre-Application Planning Meeting

- Step 1: User opens Pre-Application request form.
- Step 2: Inputs developer details and proposed development coordinates.
- Step 3: Enters meeting agenda topics and lists proposed attendees.
- Step 4: Uploads conceptual sketches and site layout plans.
- Step 5: Submits the request. No checkout fee applies for scheduling preliminary meetings.

7. Navigation Flow

NJDOT Login Page



8. Business Rules

- Milepost Validation: Proposed coordinates must fall within the physical limits of the selected Route suffix.
- No Fee Scheduling: Pre-application planning meetings do not require checkout payment.
- Authorization Certificate: If the applicant company is not the property owner of record, an Authorization Certificate signed by the owner is mandatory.
- Calculated Road Surface Fees: Opening dimensions (Length x Width) automatically dictate the fee structure based on physical square footage.

9. Required Fields & Validations

Form Section	Field Name	Type	Validation Criteria
Owner Info	Lot Owner Company	String	Mandatory; must match business registry.
Owner Info	Contact Phone	Numeric	Must be formatted as (XXX)-XXX-XXXX.
Location	Route	Dropdown	Must select from active state highway routes.
Location	Block / Lot	String	Mandatory; alphanumeric permitted.
Dimensions	Length / Width	Numeric	Positive values required; dictates fee.
Attachments	Site Plans	Upload	Mandatory; PDF format recommended.
Checkout	Card Number	Numeric	16-digit valid credit card string.
Checkout	CVV / Expiry	Numeric	3-digit CVV, year must be in the future.

10. Status Flow & Lifecycle

- Draft: Form created but payment has not been successfully processed.
- Submitted / Paid: Fee paid; visible to NJDOT staff.
- Under Review: Assigned to an engineering analyst for design check.
- Action Required: Files missing or revisions needed; returned to applicant's Action Items grid.
- Approved / Issued: Final permit generated and released as a PDF.
- Rejected: Spacing safety issues or local zoning denials; file closed.

11. Approval Workflow

1. Customer submits application and pays fee.
2. System conducts automated spatial GIS routing checks.
3. NJDOT Engineer reviews design plans and drainage metrics.
4. Inspector coordinates a physical roadside field check.
5. Final sign-off by Section Manager; permit is issued digitally.

12. Data Requirements

- Numeric Precision: Opening sizes, pipe runoff coefficients, and mileposts must be decimal values.
- Date Stamps: Dates are validated and formatted exclusively as MM/DD/YYYY.
- File Upload Limits: Enforces maximum size limits per upload and accepts only safe extensions (PDF, DOC, DOCX, JPG, PNG).

13. UI Components & Controls

- Kendo UI Grid: Used for Action Items, Permit History, and Payment Activity grids. Supports sorting, server-side data loading, and pagination.
- Kendo DatePicker: Custom date selector. Input fields are updated directly via JavaScript to bypass flaky calendar popup clicks.
- Kendo NumericTextBox: Restricts coordinate or dimension fields to numbers.
- Kendo PanelBar: Handles expandable navigation menus.

14. Document Upload/Download Workflow

Upload Flow:

- The user drags a file into the Kendo Upload dropzone or clicks to select a local file.
- An asynchronous POST request uploads the file to a secure temporary directory.
- Upon completion, the system shows a green checkmark icon (`.k-file-success` or `.k-i-check`).

Download Flow:

- Exporting grids (Excel/CSV) triggers a backend query which bundles data into a file stream and initiates a browser download.
- Accessing generated permits or invoices opens an inline popup window with authentication cookies carried over automatically.

15. Search & Grid Functionality

- Search Criteria: Standard search boxes filter Kendo grids instantly based on record IDs or addresses.
- Pagination: Handles large datasets via fast pagination controls.
- Export Utility: Extracts the active grid view into a spreadsheet.

16. Integration Points

- NJ Common Checkout: Redirects user payments to New Jersey's state financial infrastructure.
- NJDOT GIS/Map Services: Automatically validates route coordinates and setbacks.

17. Error Handling & Validation Rules

- Mandatory fields: Empty values trigger immediate red borders and a top-level alert summary panel.
- Incorrect types: Entering letters into numeric boxes is rejected. Dates must fit the MM/DD/YYYY format.
- Invalid credentials: Failed logins display a modal pop-up error: 'You have entered an invalid email address or password.'

18. Automation Coverage Summary

The customer portal is automated with a Python-based Playwright framework using Page Object Models (POMs) and Pytest. The suite supports parallel execution using xdist and incorporates cross-process file locks to prevent database conflicts during authentication.

19. Test Scenarios Covered

Test Scenario	Verified Flow	Expected Outcome
test_login_page_api_validation	Validates login page inputs.	Email/password mandatory checks pass.
test_utility_opening	Submit road openings and complete payment.	Dimensions calculated, pays successfully.
test_drainage	Submit drainage connection and runoff data.	Hydrology checks complete, redirects checkout.
test_erection_pole	Submit utility pole erection request.	Spacing details filled, saves successfully.
test_highway_occupancy	Request temporary highway closures.	Staging details submitted, paid successfully.
test_letter_interest	Submit access pre-planning inquiry.	Saves successfully without checkout fee.
test_lot_consolidation	Submit lot consolidation mapping.	Saves maps and routes to payment gateway.
test_major_planning	Submit complex subdivision details.	Spacing grids completed, paid successfully.
test_pre_application_meeting	Request pre-application planning session.	Meeting agenda saved without checkout fee.
test_street_intersection	Request connection of street intersections.	Saves layouts and routes to payment gateway.
test_payment_activity_filters	Filter activity by multiple timeframes and export.	Verifies grid updates dynamically and exports CSV.

20. Gaps & Missing Coverage

- Attachment Limit boundaries: Uploading extremely large plans or multiple document sets.
- Conditional Fields: Verification of validations when changing permit choices mid-entry.

21. Known Limitations

- AJAX Latency: Kendo loading masks sometimes cause flaky clicks under server load, resolved by explicit hidden-mask wait conditions.
- Zoom Layout Shifts: Elements shift under different browser zoom settings, requiring force-clicks on certain checkbox labels.

22. Team Knowledge Base

- Auth Cache: State files are cached inside '.auth/auth_state.json'. Clear this to trigger fresh logins.
- Staggered delays: Workers delay starting by 4-second intervals to avoid overloading the test server.

23. AI Learning Notes

- Kendo DatePicker direct set: The calendar picker UI is highly unstable under automation. Set values directly via window.jQuery/JavaScript evaluation.

24. Future Automation Opportunities

- E2E Integration: Linking customer submission with Staff Portal approval steps to test full regulatory workflows.

25. Future Product Enhancement Opportunities

- GIS coordinate selector: An interactive pin-drop map interface replacing manual milepost inputs.
- Autofill Company Profiles: Automatic registry lookup using corporate tax registration databases.